

Bridging Lean and the LMFDB

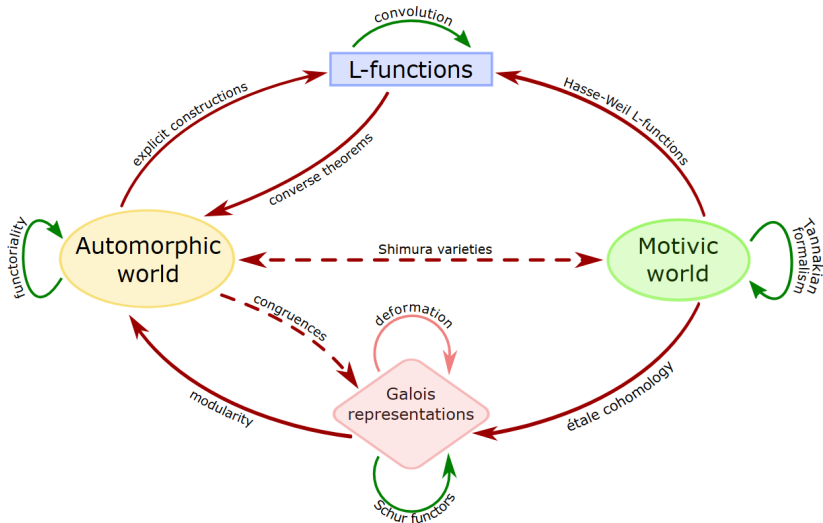
Thematic Program on AI and Mathematics
Formalization, Theorem Proving and Reasoning

David Kurniadi Angdinata

University of East Anglia

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The L-functions and modular forms database



Objects in the LMFDB

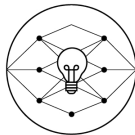
The LMFDB contains over 100 million objects with unique labels:

- 24m L-functions (rational and otherwise)
- 850k modular forms (classical, Hilbert, Bianchi, Maass, and *Siegel*)
- 42m varieties (elliptic curves over $\mathbb{Q}$ and over number fields, genus two and higher genus curves over $\mathbb{Q}$, and abelian varieties over $\mathbb{F}_q$), Belyi maps, and *modular curves*
- 21m number fields and p -adic fields
- 1m abstract groups, Galois groups, Sato–Tate groups, and *lattices*
- 31m Dirichlet characters and Artin representations
- *60k hypergeometric motives*

There are also completeness and reliability considerations for each family.

Each object has its own page, with data from computer algebra systems, and with links to the **knowledge** database and other related objects.

Scalable theorem proving via mathematical databases



Chris Birkbeck



David Roe



Andrew Sutherland

Objectives of the project

The project has many closely-related objectives.

On the side of the LMFDB:

- link the knowledge database to definitions in `mathlib`
- add a command to preview the corresponding Lean statements
- generate Lean proofs for simple or certifiable computations

On the side of Lean and `mathlib`:

- add attributes to definitions that have LMFDB knows
- extend the property testing tactic to LMFDB objects
- prove results assuming completeness and reliability of the LMFDB

The LeanBridge repository hosts a rough blueprint, mostly on definitions for number fields, elliptic curves over $\mathbb{Q}$, and classical modular forms.

Workshop in Norwich

A workshop with over 50 participants was held in the University of East Anglia from Monday, 29 June 2026 to Friday, 3 July 2026.



There will be another workshop in the Massachusetts Institute of Technology from Monday, 25 January 2027 to Friday, 29 January 2027.

Number fields

A large group worked on certifying the class number formula

$$\lim_{s \rightarrow 1} ((s-1) \cdot \zeta_K(s)) = \frac{2^{r_1} \cdot (2\pi)^{r_2} \cdot \text{Reg}_K \cdot h_K}{w_K \cdot \sqrt{|D_K|}},$$

which was formalised in `mathlib` by Xavier Roblot in May 2025.

- *Anne Baanen*, *Sander Dahmen*, and *Alain Chavarri Villarello* had previously certified many invariants of $\mathcal{O}_K$, including D_K and h_K .
- *David Ledvinka* had a tactic for interval arithmetic.
- *Bhavik Mehta* and *Heather Macbeth* had previous results on isolation of roots and error bounds of matrix determinants.
- *Chris Birkbeck* computed an error bound for $\lim_{s \rightarrow 1} ((s-1) \cdot \zeta_K(s))$.
- Along with everyone else above, *Barinder Banwait*, *Thomas Browning*, *Stephan Elsenhans*, *Xavier G  n  reux*, *Artie Khovanov*, and *Arend Mellendijk* computed an error bound for Reg_K .

They verified the class number formula for one sextic number field.

Elliptic curves

In contrast, `mathlib` has little on elliptic curves, mostly formalised by Junyan Xu and me. This includes the group law, the basics of division polynomials, and parts of the multiplication-by- n formula.

- Chris Birkbeck and I formalised the Lutz–Nagell theorem, with ongoing work to compute torsion subgroups over $\mathbb{Q}$.
- *Michael Stoll* formalised heights over number fields, with potentially ongoing work on the Mordell–Weil theorem.
- Riccardo Brasca, Richard Hill, and Edison Xie continued their work on continuous cohomology and defined the Tate–Shafarevich group.
- Bhavik Mehta certified lower rank bounds, including the rank ≥ 29 Elkies–Klagsbrun curve discovered in 2024, and even found an error in the rank ≥ 21 Nagao–Kouya curve discovered in 1994.
- Håvard Damm-Johnsen, Zachary Feng, and Jane Shi added the Lean command for curve and isogeny class pages in the LMFDB.

Modular forms

Similarly, `mathlib` has little on modular forms, mostly formalised by Chris Birkbeck and David Loeffler. This includes actions on $\mathbb{H}$, classical Eisenstein series, q -expansions, and finite dimensionality for $SL_2(\mathbb{Z})$.

- Chris Birkbeck maintains the massive `AINTLib/LeanModularForms` repository, which contains the valence formula for $SL_2(\mathbb{Z})$ and the strong multiplicity one theorem for newforms with nebentypus.
- David Loeffler continued his work on L-functions for modular forms.
- Ramin Takloo-Bighash and Samuel Yin formalised duality for cusp forms with nebentypus and coefficient fields of Hecke eigenforms.
- Wenrong Zou and James Kiln formalised Eisenstein series with nebentypus and computed their q -expansions.
- William Coram, John Cremona, Nicola Falciola, Sayeb Salam, and Samuel Yin worked on uniformisation theorems over $\mathbb{C}$ and $\mathbb{Q}_p$.

Miscellaneous

Numerous other projects!

- Bhavik Mehta developed the lookup tactic for Galois groups, number fields, elliptic curves, and modular forms.
- *Mario Carneiro*, *James Davenport*, and Michail Karatarakis developed kernel-reducible computable multivariate polynomials.
- Yunfei Li and Jiaqi Wang worked on an interface for p -adic fields and their ramification theory for finite extensions.
- Peiran Wu formalised Gassmann equivalences of subgroups.
- Justus Springer and Peiran Wu worked on affine group schemes.
- Ammara Ansari, Giovanna De Lauri, Vaishnav Nambiar, Yufei Qian, Pratyush Venkatakrishnan, and Xindi Yang worked on lattices and kissing numbers, supervised by Riccardo Brasca and Kevin Buzzard.